Answers on
Booklet - 9

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a)(i)	stress = force / (cross sectional) area	B1
(ii)	strain = extension / (original) length	B1
(b) (i)	elastic returns to its original length/shape when the force/load is removed	B1
	plastic does not regain its original length/size when the load is removed	B1
	(allow ½ if removal of load is not specified but remainder is clear)	
(ii)	force/stress/strain/extension beyond which the material does not return to its original length (when the load is removed)	B1 B1
	(point beyond….scores one only)	
(c)(i)	unit: Pa / N m^{-2}	B1
(ii)	Any <u>seven</u> from:	
	• Appropriate arrangement drawn	B1
	• Measure length	B1
	• Apply force and measure the extension	B1
	• Take a series of readings	B1
	• Measure the diameter	B1
	• Graph of stress against strain / force-extension graph	B1
	• Gradient for E / E = stress/strain / Gradient = $EA/_L$	B1
	• Point of detail e.g. second wire, micrometer used for diameter, Vernier for extension, very long wire	B1

2.

i	momentum before= 2mv - 3mv = - mv	1
	momentum after = 4mv – m5v = - mv	1
Ii	initial k.e. = 2.5 mv^2; final k.e. = 14.5 mv^2 (so Δk.e. = 12 mv^2)	2
iii	k.e. of n is 12.5 mv^2 ; so takes all of the extra k.e./AW	2
iv	12 mv^2 = 12 x 1.67 x 10^{-27} x 1.19^2 x 10^{14} = 2.84 x 10^{-12} (J) *ecf ii*	1

3.

a	**i**	3.8 ± 0.3 (N s)	**1**
	ii	momentum (of the ball) *accept impulse*	**1**
	iii	mv = 3.8 or v = 3.8/0.16; = 23 (m s^{-1}) *ecf a*	**2**
	iv	use F = ma giving 24 = 0.16a; a = 150 (m s^{-2})	**2**

4.

a		mgh or 0.014 x 9.8 x h ;	**1**
		= 0.22 J so h = 1.6 m	**1**
b		k.e. increases;	**1**
		at a decreasing rate	**1**
c	**i**	½ mv^2 = 0.11	**1**
		v = √(0.11 / 0.007) = (3.96 m s^{-1})	**1**
	ii	F = mg = 0.014 x 9.8 = 0.137 or 0.14 (N)	**1**
	iii	Fv = mgv or 0.14 x 4.0	**1**
		= 0.54 to 0.56 (W) *ecf c(ii)*	**1**
d		k.e. after bounce = 0.088 J giving u = √(0.088/0.007) = 3.55 m	**1**
		s^{-1}	**1**
		change in momentum = m (v ± u)	**1**
		= 0.014(7.55)= 0.11;	**1**
		kg m s^{-1} /N s	

5.

V units: m^3 (*allow metres cubed or cubic metres*)

Pressure units: kg m s^{-2} / m^2 (*allow use of* $P = \rho gh$)
Units: kg m^{-1} s^{-2}

6.

speed : **decreases**	B1
frequency: **stays the same**	B1
wavelength : **decreases**	B1

7.

(a)	**plane wavefronts** approaching a gap	B1	
	'semicircular' (allow ANY spreading shape) wavefronts leaving the gap	B1	
	no change of wavelength stated or shown (generously judged by eye)	B1	[3]
(b)	sound waves are **diffracted**	B1	
	around the wall (OR spread out on other side of wall)	B1	[2]
	{Allow 1 mark for "sound is transmitted through the wall but light is not" OR "sound λ>>light λ and hence no diffraction of light"}		

8.

(a)(i)	correct sketch: **same shape** shifted to the **right**	B1	[1]
	At least one full cycle must be drawn		
(ii)	arrows showing X↑, Z↓	B1 + B1	[2]
	Award marks as shown irrespective of wave drawn by cand. but look for the new positions of X and Z shown on the cand's new wave position.		
(iii)	wavelength = 2 x 0.24 = 0.48m	B1	
	recall of v=fλ	C1	
	v = 3.6 x 0.48 = **1.73** m s^{-1}	A1	[3]
	Allow ecf for candidate's value of λ; eg λ= 0.24m v = 0.865 m/s scores 2 marks		
(iv)	1. wave speed **NO CHANGE**	B1	
	2. wavelength is shortened	C1	
	to a new value of **0.24 m OR is halved**	A1	[3]
	{allow ecf for cand's value from (iii)}		

9.

(a) (i) a progressive wave transfers energy (WTTE) B1 [1]
{allow "wave profile moves through space" OR crest/troughs move along the medium}
{allow "waves that move from one place to another" but not "waves move"}

(ii) longitudinal: vibrations/motions PARALLEL (to wave direction) B1
{allow back and forth}
transverse: vibrations/motions PERPENDICULAR (to wave direction) - B1 [2]

(b) a straight object vibrated in the water (WTTE) B1

REDUCE FREQUENCY of wave source B1
{allow 'increase the depth of the water'}
REDUCE DEPTH of water in the tank B1 [3]

10.

(a) **(i)** Motion in random directions; (1)
with a range of speeds; (1)
and changes of direction on collision with metal atoms. (1) [3]

(ii) Random motion with varying speeds continues; (1)
Electrons accelerate (between collisions with atoms); (1)
in opposite direction to the current / towards the positive terminal or wtte; (1) [3]

(b) **(i)** Mean / average velocity / rate of change of displacement (along the wire). [1]

(ii) Sketch of cylinder with length labelled OR reference to specific length of wire; (1)
Total charge of free electrons in this cylinder / length = e.g. nALe (1)
I = Q / t stated and used to give I = nAve (1) [3]

11.

(a)(i) Correct direction shown (anticlockwise) B1

(a)(ii) Direction in which positive charges / ions move /
Direction / flow / current / from positive to negative /
Flow of (positive) charge from positive to negative /
Direction / flow opposite to electron flow B1

(a)(iii) $Q = It$ (Allow any subject with or without delta notation)
C1

$$I = \frac{0.76}{5.0 \times 60}$$ C1

current = 2.53×10^{-3} (A) ≈ 2.5×10^{-3} (A) A1
(0.152 / 0.15 (A) scores 1/3)

12.

a	**i**	travel through a vacuum	B1	**allow** travel at c (in a vacuum)
b	**ii**	A gamma; C uv; F microwave	B3	**allow** 1 mark for A radio; C ir; F X-ray

13.

amplitude = 1.5 mm

wavelength = 25 / 6
= 4.2 cm or 4.2×10^{-2} m

$v = \lambda / T$ or $v = f\lambda$ <u>and</u> $T = 1/f$
$T = 4.2 / 7.5 = 0.56$ s

14.

(a) any **two** valid points: e.g
in longitudinal waves the vibrations are parallel to wave direction (WTTE) }
in transverse waves the vibrations are perpendicular to wave direction (WTTE) }
transverse waves <u>can</u> be polarised (OR longitudinal waves are not be polarised) }
(all) longitudinal waves need a medium } B1+B1 [2]

(b) (i) vibrations "V" correctly labelled OR (NOT) B1 [1]
(ii) compression "C" correctly shown anywhere on the spring B1 [1]
(iii) wavelength "λ" correctly shown: e.g. between neighbouring compressions B1 [1]

(generously judged: i.e. somewhere between 28 and 34 mm)

(c) wavelength REDUCES B1
because $v=f\lambda$ AND v remains constant (WTTE) B1 [2]